Tutorial 6 solutions

Question 1 - Joint PMF

The joint PMF is

$$p(x,y) = c(x+y+1), \quad x,y \in \{0,1,2\}.$$

(a) Find c

A valid PMF must sum to 1:

$$\sum_{x=0}^{2}\sum_{y=0}^{2} c\,(x+y+1) = 1.$$

Now

$$\sum_{x=0}^{2}\sum_{y=0}^{2}(x+y+1) = 9 + 3(0+1+2) + 3(0+1+2) = 27.$$

Hence

$$27c = 1 \Longrightarrow c = \frac{1}{27}.$$

(b) Compute $P(X+Y \le 2)$

The allowed pairs are

$$(0,0),(0,1),(1,0),(0,2),(1,1),(2,0).$$

Therefore

$$P(X+Y \le 2) = \frac{1}{27}[1+2+2+3+3+3] = \frac{14}{27} \approx 0.5185.$$

(c) Test independence with one explicit check

We do not need all marginals. Compute only what is needed for $(0,0)$:

$$P(X=0) = \frac{1+2+3}{27} = \frac{2}{9}, \qquad P(Y=0) = \frac{2}{9}.$$

But

$$p(0,0) = \frac{1}{27} = \frac{3}{81}, \qquad P(X=0)P(Y=0) = \frac{4}{81}.$$

Since these are unequal,

X and Y are not independent.

Question 2 - Joint PDF

The density is

$$f(x,y) = cxy, \quad 0 < y < x < 1.$$

(a) Find c

Normalize over the triangular support:

$$1 = \int_0^1 \int_0^x c\,xy\,dy\,dx = \int_0^1 c\,x\left[\frac{y^2}{2}\right]_0^x dx = \frac{c}{2}\int_0^1 x^3\,dx = \frac{c}{8}.$$

Thus

$$c = 8.$$

(b) Compute $P(Y > X/2)$

Inside the original support, $0 < y < x$. Adding $Y > X/2$ gives

$$0 < x < 1, \qquad \frac{x}{2} < y < x.$$

Hence

$$P(Y > X/2) = \int_0^1 \int_{x/2}^x 8\,xy\,dy\,dx = \int_0^1 4\,x\left[x^2 - \frac{x^2}{4}\right]dx = \int_0^1 3\,x^3\,dx = \frac{3}{4} = 0.75.$$

(c) Why X and Y cannot be independent from the support

The support is triangular: $0 < y < x < 1$. It is not a Cartesian product of an x-interval and a y-interval. For example, values of x and y may each lie in their marginal ranges while the pair (x, y) is impossible whenever $y \geq x$. Therefore the possible values of Y depend on X, so

X and Y cannot be independent.

Question 3 - Marginal Distributions

The joint density is

$$f(x,y) = 3(x+y), \quad x > 0,\ y > 0,\ x + y < 1.$$

(a) Derive $f_X(x)$

For a fixed x, the condition $x + y < 1$ gives $0 < y < 1 - x$. Thus

$$f_X(x) = \int_0^{1-x} 3\,(x+y)\,dy = 3\left[xy + \frac{y^2}{2}\right]_0^{1-x} = 3\left[x(1-x) + \frac{(1-x)^2}{2}\right] = \frac{3}{2}(1-x^2), \quad 0 < x < 1.$$

Outside this interval, $f_X(x) = 0$.

By symmetry,

$$f_Y(y) = \frac{3}{2}(1-y^2), \quad 0 < y < 1.$$

(b) Compute $P(X > 0.5)$

$$P(X > 0.5) = \int_{1/2}^{1} \frac{3}{2}(1-x^2)\,dx = \frac{3}{2}\left[x - \frac{x^3}{3}\right]_{1/2}^{1} = \frac{5}{16} = 0.3125.$$

(c) Determine whether X and Y are independent

Take $x = y = 1/4$, which lies inside the joint support. Then

$$f\left(\frac{1}{4}, \frac{1}{4}\right) = 3\left(\frac{1}{4} + \frac{1}{4}\right) = \frac{3}{2}.$$

Also

$$f_X\left(\frac{1}{4}\right) = f_Y\left(\frac{1}{4}\right) = \frac{3}{2}\left(1 - \frac{1}{16}\right) = \frac{45}{32}.$$

Therefore

$$f_X\left(\frac{1}{4}\right) f_Y\left(\frac{1}{4}\right) = \left(\frac{45}{32}\right)^2 = \frac{2025}{1024} \approx 1.9775 \neq 1.5.$$

Hence

X and Y are not independent.

Question 4 - Moments and Probability Density Functions

The eight observations are:

Observation	Sport	Cruiser	Off-road
1	170	168	100
2	175	172	105
3	180	176	110

Observation	Sport	Cruiser	Off-road
4	185	180	115
5	190	184	120
6	178	170	108
7	182	178	112
8	176	174	106

For every category,

$$\mu_1' = \frac{1}{n}\sum x_i, \qquad \mu_2' = \frac{1}{n}\sum x_i^2, \qquad \mu_1 = 0,$$

and

$$\mu_2 = \sigma^2 = \mu_2' - (\mu_1')^2.$$

Sport bikes

$$\sum x_i = 1436, \qquad \sum x_i^2 = 258034.$$

Therefore

$$\mu_1' = \mu = \frac{1436}{8} = 179.5, \qquad \mu_2' = \frac{258034}{8} = 32254.25.$$

The first central moment is

$$\mu_1 = \frac{1}{8}\sum(x_i - 179.5) = 0.$$

The variance is

$$\sigma^2 = 32254.25 - (179.5)^2 = 34, \qquad \sigma = \sqrt{34} = 5.83095.$$

Cruisers

$$\sum x_i = 1402, \qquad \sum x_i^2 = 245900.$$

Hence

$$\mu_1' = \mu = 175.25, \qquad \mu_2' = 30737.5,$$

$$\mu_1 = 0, \qquad \sigma^2 = 30737.5 - (175.25)^2 = 24.9375, \qquad \sigma = \sqrt{24.9375} = 4.99375.$$

Off-road bikes

$$\sum x_i = 876, \qquad \sum x_i^2 = 96194.$$

Hence

$$\mu_1' = \mu = 109.5, \qquad \mu_2' = 12024.25,$$

$$\mu_1 = 0, \qquad \sigma^2 = 12024.25 - (109.5)^2 = 34, \qquad \sigma = \sqrt{34} = 5.83095.$$

Summary

Category	Mean μ	Second raw moment μ_2'	Variance σ^2	SD σ
Sport	179.5000	32254.2500	34.0000	5.8310
Cruiser	175.2500	30737.5000	24.9375	4.9937
Off-road	109.5000	12024.2500	34.0000	5.8310

This verifies numerically that

$$\mu_1' = \mu, \qquad \mu_1 = 0, \qquad \sigma^2 = \mu_2 = \mu_2' - (\mu_1')^2.$$

Explicit Normal PDFs

Using

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right],$$

we obtain

$$f_S(x) = \frac{1}{\sqrt{34}\sqrt{2\pi}} \exp\left[-\frac{(x-179.5)^2}{68}\right]$$

$$f_C(x) = \frac{1}{\sqrt{24.9375}\sqrt{2\pi}} \exp\left[-\frac{(x-175.25)^2}{49.875}\right]$$

$$f_O(x) = \frac{1}{\sqrt{34}\sqrt{2\pi}} \exp\left[-\frac{(x-109.5)^2}{68}\right]$$

Numerically, the leading constants are approximately 0.068418, 0.079888, and 0.068418, respectively.

Question 5 - Computational Generation and Comparison of PDF Curves

A direct Python implementation is:

```
import numpy as np
import matplotlib.pyplot as plt

mu_s, var_s = 179.5, 34.0
mu_c, var_c = 175.25, 24.9375
mu_o, var_o = 109.5, 34.0

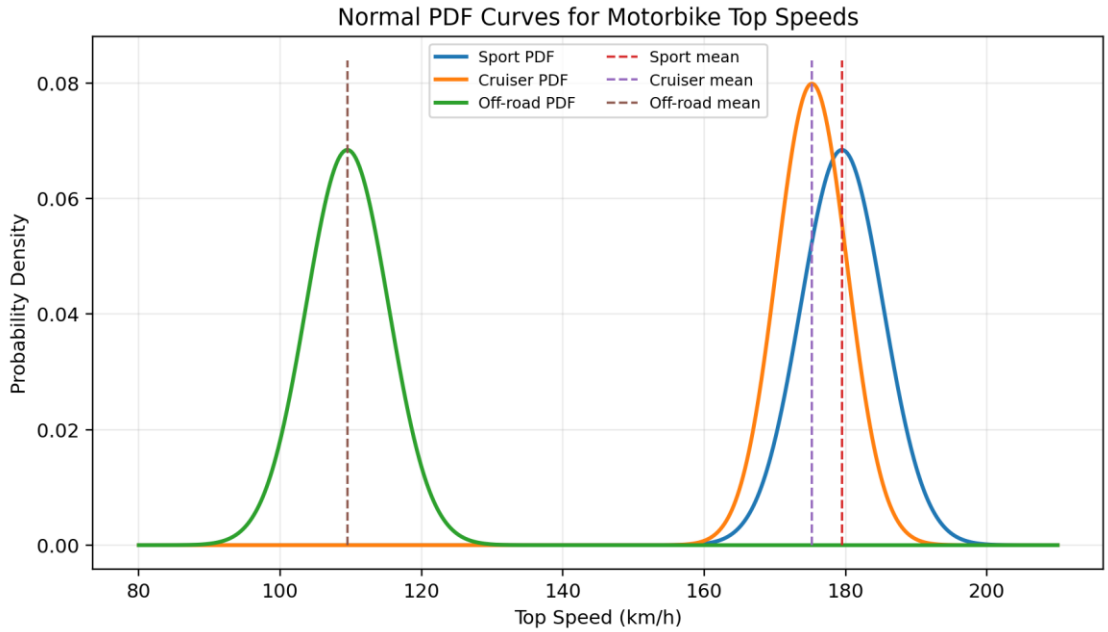
sigma_s = np.sqrt(var_s)
sigma_c = np.sqrt(var_c)
sigma_o = np.sqrt(var_o)
```

```
x = np.linspace(80, 210, 1000)

def normal_pdf(x, mu, sigma):
    return (1 / (sigma * np.sqrt(2*np.pi))) * \
        np.exp(-((x - mu)**2) / (2*sigma**2))

fS = normal_pdf(x, mu_s, sigma_s)
fC = normal_pdf(x, mu_c, sigma_c)
fO = normal_pdf(x, mu_o, sigma_o)

plt.plot(x, fS, label="Sport")
plt.plot(x, fC, label="Cruiser")
plt.plot(x, fO, label="Off-road")
plt.axvline(mu_s, linestyle="--", label="Sport mean")
plt.axvline(mu_c, linestyle="--", label="Cruiser mean")
plt.axvline(mu_o, linestyle="--", label="Off-road mean")
plt.xlabel("Top Speed (km/h)")
plt.ylabel("Probability Density")
plt.title("Normal PDF Curves for Motorbike Top Speeds")
plt.legend()
plt.show()
```



PDF curves obtained from the parameters in Question 4.

Graphical interpretation

1. **Most similar curves:** Sport and Cruiser. Their means are only $179.5 - 175.25 = 4.25$ km/h apart, and their spreads are also similar.
2. **Clearly separated curve:** Off-road. Its mean is 109.5 km/h, far below the other two means.

3. **Effect of means:** The mean sets the horizontal center of each bell curve. Larger means shift a curve to the right; smaller means shift it to the left.
4. **Effect of variances:** Sport and Off-road both have variance 34, so they have the same width and peak height. Cruiser has the smaller variance 24.9375, so its curve is slightly narrower and taller.
5. **Greatest visual overlap:** Sport and Cruiser.

Question 6 - Covariance and Correlation Between the PDF Curves

The revised question supplies the sampled PDF values directly:

Top speed x	Sport S	Cruiser C	Off-road O
90	0.000000	0.000000	0.000255
105	0.000000	0.000000	0.050797
120	0.000000	0.000000	0.013522
135	0.000000	0.000000	0.000005
150	0.000000	0.000000	0.000000
165	0.003107	0.009719	0.000000
180	0.068167	0.050818	0.000000
195	0.001999	0.000032	0.000000

These columns are treated as three numerical vectors of length $m = 8$. They represent PDF heights at the same speeds, not the original bike-speed observations.

(a) Mean PDF value of each vector

For Sport,

$$\bar{S} = \frac{0 + 0 + 0 + 0 + 0 + 0.003107 + 0.068167 + 0.001999}{8} = 0.009159125.$$

For Cruiser,

$$\bar{C} = \frac{0 + 0 + 0 + 0 + 0 + 0.009719 + 0.050818 + 0.000032}{8} = 0.007571125.$$

For Off-road,

$$\bar{O} = \frac{0.000255 + 0.050797 + 0.013522 + 0.000005}{8} = 0.008072375.$$

(b) Population covariances

The formula is

$$\mathrm{Cov}(F, G) = \frac{1}{8}\sum_{j=1}^{8}\left(F_j - \bar{F}\right)\left(G_j - \bar{G}\right).$$

For S and C, the complete deviation-product calculation is:

x	$S_j - \bar{S}$	$C_j - \bar{C}$	Product
90	-0.009159125	-0.007571125	6.9344880×10^{-5}
105	-0.009159125	-0.007571125	6.9344880×10^{-5}
120	-0.009159125	-0.007571125	6.9344880×10^{-5}
135	-0.009159125	-0.007571125	6.9344880×10^{-5}
150	-0.009159125	-0.007571125	6.9344880×10^{-5}
165	-0.006052125	0.002147875	$-1.2999208 \times 10^{-5}$
180	0.059007875	0.043246875	2.5519062×10^{-3}
195	-0.007160125	-0.007539125	5.3981077×10^{-5}
Sum of products			**0.002939612465**

Thus

$$\mathrm{Cov}(S, C) = \frac{0.002939612465}{8} = 0.0003674516.$$

Applying the same formula to the other pairs gives

$$\mathrm{Cov}(S, O) = -0.0000739359$$

and

$$\mathrm{Cov}(C, O) = -0.0000611170.$$

(c) Standard deviations and correlation coefficients

Using the population standard deviation

$$\sigma_F = \sqrt{\frac{1}{8}\sum_{j=1}^{8}\left(F_j - \bar{F}\right)^2},$$

we obtain

$$\sigma_S = 0.02233068, \qquad \sigma_C = 0.01665215, \qquad \sigma_O = 0.01674044.$$

Now

$$r_{FG} = \frac{\mathrm{Cov}(F, G)}{\sigma_F \sigma_G}.$$

Therefore

$$r_{SC} = \frac{0.0003674516}{(0.02233068)(0.01665215)} = 0.9882,$$

$$r_{SO} = \frac{-0.0000739359}{(0.02233068)(0.01674044)} = -0.1978,$$

$$r_{CO} = \frac{-0.0000611170}{(0.01665215)(0.01674044)} = -0.2192.$$

(d) Interpretation

1. **Most similar PDF curves:** Sport and Cruiser, because $r_{SC} \approx 0.9882$ is very close to $+1$.
2. **Least similar curve:** Off-road is weakly negatively correlated with both Sport and Cruiser on this sampled grid (-0.1978 and -0.2192).
3. **Agreement with Question 5:** Yes. Sport and Cruiser overlap strongly in the graph, while the Off-road curve is centered much farther to the left.
4. **Why the same x-values are essential:** Covariance and correlation compare pairs (F_j, G_j). Each pair must refer to the same physical top speed. If different x-grids were used, the calculation would pair PDF heights from unrelated locations and would no longer measure how the two curves rise and fall together across the speed axis.

Important interpretation: These correlation coefficients measure similarity between the sampled PDF heights. They are not correlations between the original Sport, Cruiser, and Off-road top-speed random variables.